12B 1 LATTICE ENERGY, $\Delta_{LE}H$ AND BORN–HABER CYCLES

SPECIFICATION REFERENCE
12.12(i) 12.12(ii) 12.12(iii) 12.13 12.18 PART

LEARNING OBJECTIVES

- Be able to define the terms:
  (i) standard enthalpy change of atomisation, $\Delta_{at}H$
  (ii) electron affinity
  (iii) lattice energy (as the exothermic process for the formation of one mole of an ionic solid from its gaseous ions).
- Understand that lattice energy provides a measure of the strength of ionic bonding.
- Construct Born–Haber cycles and carry out related calculations.
- Understand the effect of ionic charge and ionic radius on the value of the lattice energy of an ionic compound.

LATTICE ENERGY

In **Book 1**, we saw that bond enthalpies can be used as a measure of the strength of the covalent bonding in molecules. The equivalent energy change for ionic bonding in ionic compounds is lattice energy, $\Delta_{LE}H$ (or lattice enthalpy).

The lattice energy of a compound is the energy change when one mole of the ionic solid is formed from its gaseous ions. If standard conditions of 100 kPa and a stated temperature (usually 298 K) are applied (as shown by $^{\ominus}$), then the energy change is called the **standard lattice energy**.

The equation that represents the standard lattice energy of sodium chloride is:

$$Na^+(g) + Cl^-(g) \rightarrow NaCl(s) \qquad \Delta_{LE}H^{\ominus} = -780\,kJ\,mol^{-1}$$

For magnesium chloride it is:

$$Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s) \qquad \Delta_{LE}H^{\ominus} = -2526\,kJ\,mol^{-1}$$

LEARNING TIP

The terms 'lattice energy' and 'lattice enthalpy' are commonly used as if they mean exactly the same thing. You will often find both terms used within the same textbook, article or website, including some university websites.

In fact, there is a difference between them. It relates to the conditions under which they are calculated. However, the difference is negligible when compared with the different values for lattice energies that you will find in different sources.

Unless you go on to study chemistry at degree level, you do not need to worry about the difference between the two terms.

DID YOU KNOW?

There are two ways of defining lattice energy. The one mentioned in the specification is the one we have already defined. This is sometimes called lattice energy of *formation*, since the compound is being formed from its ions. Using this definition, the energy change will always be negative.
However, some books define it as lattice energy of *dissociation*, in which it is the energy change when one mole of the compound is broken down (or dissociated) into its ions. In this case, the energy change will be positive.

FACTORS AFFECTING THE MAGNITUDE OF LATTICE ENERGY

The lattice energy of magnesium chloride, $Mg^{2+}(Cl^-)_2$, is much larger (i.e. more negative) than that of sodium chloride, Na^+Cl^-. A number of factors are responsible for this difference:

- The first thing to realise is that a magnesium ion carries twice the charge of a sodium ion.
- Second, there are more cation-to-anion interactions in magnesium chloride because there are twice as many chloride ions per cation than in sodium chloride.
- A third factor is the distance between the centres of the cations and their neighbouring anions, which is equal to the sum of their ionic radii. This distance is determined in part by the relative sizes of the ions involved (the Mg^{2+} ion is smaller than Na^+, so reducing the sum of the ionic radii) and also by the type of lattice structure the compound has. In fact, the relative ion sizes determine the type of lattice structure.

Table A shows the effects that both the inter-ionic distance and the charges on the ions have on the lattice energy.

COMPOUND	INTER-IONIC DISTANCE/nm	CHARGES ON THE IONS	LATTICE ENERGY/ kJ mol^{-1}
LiF	0.207	+1, −1	−1031
NaF	0.235	+1, −1	−918
CaF_2	0.233	+2, −1	−2630
Li_2O	0.214	+1, −2	−2814
MgO	0.212	+2, −2	−3791
Al_2O_3	0.193	+3, −2	−15 504

table A

Comparison of LiF and NaF, where the charges on the ions are the same, shows that a decrease in the distance between the centres of the two ions (the inter-ionic distance) results in a more negative value for the lattice energy.

Comparison of NaF and CaF_2, where the inter-ionic distances are almost the same, shows that an increase in charge of even one of the ions (Ca^{2+} as opposed to Na^+) results in a more negative value for the ionisation energy. Similar comparisons can be made between Li_2O and MgO and also between Li_2O and Al_2O_3.

Finally, as we will see later in this topic, there are also covalent interactions between the ions, and these affect the magnitude of the lattice energy. The values quoted in **table A** take these interactions into consideration since they are calculated values using a Born–Haber cycle (see later in this section).

LEARNING TIP

Lattice energies are determined by:

- the magnitudes of the charges on the ions
- the sum of the ionic radii
- the type of lattice structure
- the extent of covalent interactions between the ions.

At International A Level, you are not required to know the different types of lattice structures. In addition, the extent of the covalent interactions is already taken into consideration in the quoted values for lattice energy. Therefore, you need only consider the magnitude of the charges on the ions and the sizes of the ions.

STANDARD ENTHALPY CHANGE OF ATOMISATION, $\Delta_{at}H^{\ominus}$

The enthalpy change measured at a stated temperature (usually 298 K) and 100 kPa when one mole of gaseous atoms is formed from an element in its standard state is called the **standard enthalpy change of atomisation** of the element (see **Topic 6 (Book 1: IAS)**). It is given the symbol $\Delta_{at}H^{\ominus}$.

Equations representing some standard enthalpy changes of atomisation at 298 K are given below.

$C(s) \rightarrow C(g)$ $\quad \Delta_{at}H^{\ominus} = +717\,kJ\,mol^{-1}$

$Na(s) \rightarrow Na(g)$ $\quad \Delta_{at}H^{\ominus} = +107\,kJ\,mol^{-1}$

$\frac{1}{2}H_2(g) \rightarrow H(g)$ $\quad \Delta_{at}H^{\ominus} = +218\,kJ\,mol^{-1}$

$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$ $\quad \Delta_{at}H^{\ominus} = +122\,kJ\,mol^{-1}$

LEARNING TIP

Note that the standard enthalpy change of atomisation is the enthalpy change when one mole of gaseous *atoms* is formed. In the case of elements that exist as polyatomic molecules, it is *not* the enthalpy change when one mole of gaseous *molecules* is atomised.

ELECTRON AFFINITY

The first **electron affinity** of an element, 1st *EA*, is the energy change when each atom in one mole of atoms in the gaseous state gains an electron to form a −1 ion.

The equations below represent the first electron affinities of some elements.

$Cl(g) + e^- \rightarrow Cl^-(g)$ $\quad 1^{st}\,EA = -349\,kJ\,mol^{-1}$

$Br(g) + e^- \rightarrow Br^-(g)$ $\quad 1^{st}\,EA = -325\,kJ\,mol^{-1}$

$O(g) + e^- \rightarrow O^-(g)$ $\quad 1^{st}\,EA = -141\,kJ\,mol^{-1}$

The first electron affinity has a negative value for many elements, including the alkali metals. Examples include:

$Li(g) + e^- \rightarrow Li{-}(g)$ $\quad 1^{st}\,EA = -60\,kJ\,mol^{-1}$

and

$Na(g) + e^- \rightarrow Na^-(g)$ $\quad 1^{st}\,EA = -53\,kJ\,mol^{-1}$

There is a notable exception with the noble gases. For these, repulsion caused by the electrons already present in the valence shell results in a positive value for the first electron affinity as the additional electron would have to occupy a new valence shell.

In contrast, second electron affinities, 2nd *EA*, tend to be positive, as energy is required to overcome the repulsion between the negative ion and the electron being added. For example:

$O^-(g) + e^- \rightarrow O^{2-}(g)$ $\quad 2^{nd}\,EA = +798\,kJ\,mol^{-1}$

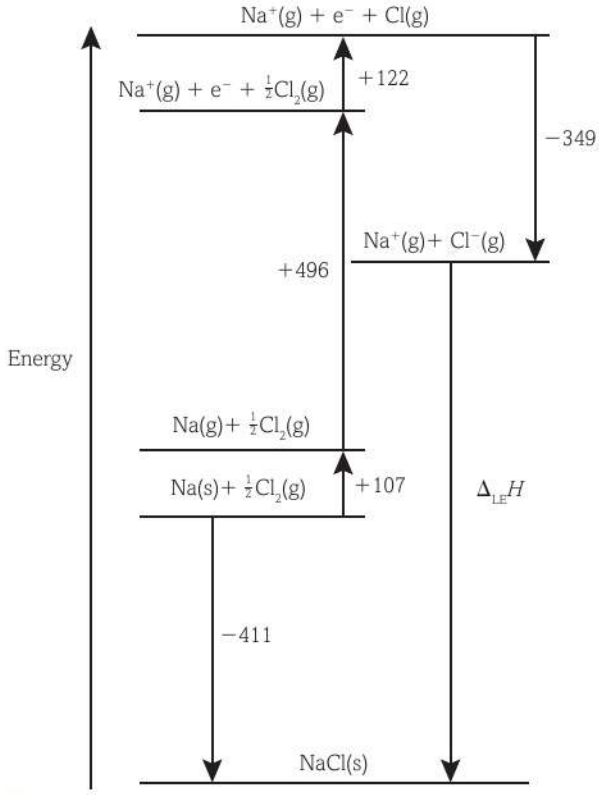
Therefore, the formation of the oxide ion, O^{2-}, in the gaseous state from its atom in the gaseous state is an endothermic process overall:

$O(g) + 2e^- \rightarrow O^{2-}(g)$ $\quad 1^{st}\,EA + 2^{nd}\,EA = +657\,kJ\,mol^{-1}$

This raises an interesting question: why does oxygen form O^{2-} ions in its binary ionic compounds rather than O^- ions, as the O^- ion would appear to be the more energetically favourable state? We will attempt to answer this question later in this topic.

BORN–HABER CYCLES

We are now in a position to consider the overall energy changes that take place when an ionic compound is made from its elements. These energy changes are summarised in an energy level diagram called the Born–Haber cycle. **Fig A** shows the Born–Haber cycle for sodium chloride.



▲ **fig A** Born–Haber cycle for sodium chloride.

EXAM HINT

Examiners may ask you to label enthalpy change of atomisation, enthalpy change of ionisation, electron affinity and enthalpy change of formation on Born–Haber cycle diagrams like **fig A**. Make sure that you can recognise each process and label it correctly.

The cycle includes the following energy changes, all of which can be determined experimentally:

- The enthalpy change of formation of NaCl(s), $\Delta_f H[NaCl(s)] = -411\,kJ\,mol^{-1}$
- $\Delta_{at}H[Na(s)] = +107\,kJ\,mol^{-1}$
- the first ionisation energy, $1^{st}\,IE[Na(g)] = +496\,kJ\,mol^{-1}$
- $\Delta_{at}H[Cl_2(g)] = +122\,kJ\,mol^{-1}$
- $EA[Cl(g)] = -349\,kJ\,mol^{-1}$

Applying Hess's Law to the cycle gives us:

$$+107 + 496 + 122 + (-349) + \Delta_{LE}H = -411$$

Hence:

$$\Delta_{LE}H[NaCl(s)] = -411 - 107 - 496 - 122 + 349\,kJ\,mol^{-1}$$
$$= -787\,kJ\,mol^{-1}$$

LEARNING TIP

Note: see **Topic 2 (Book 1: IAS)** for the definition of ionisation energy and **Topic 6 (Book 1: IAS)** for an explanation of Hess's Law.

WHY IS CALCIUM OXIDE $Ca^{2+}O^{2-}$ AND NOT $Ca^{+}O^{-}$?

You may think that the answer to this question is obvious. You might say that calcium oxide is $Ca^{2+}O^{2-}$ because both ions would then have a stable noble gas electronic configuration.

If that is what you believe, then the following may come as a surprise. Acquiring a noble gas electronic configuration is not a reason for electronic changes to take place when atoms and molecules react together.

After all, many ions exist in stable compounds, in particular, cations of the transition metals, in which they do not have the electronic configuration of a noble gas. In fact, compounds in which the ions do have a noble gas electronic configuration are in the minority.

So, why does calcium oxide form as $Ca^{2+}O^{2-}$ and not as $Ca^{+}O^{-}$, and why are we asking the question?

You may remember that we have mentioned that the formation of $O^-(g)$ from O(g) is exothermic:

$$O(g) + e^- \rightarrow O^-(g) \qquad 1^{st}\,EA = -141\,kJ\,mol^{-1}$$

However, the formation of $O^{2-}(g)$ from O(g) is endothermic:

$$O(g) + 2e^- \rightarrow O^{2-}(g) \qquad 1^{st}\,EA + 1^{st}\,EA = +657\,kJ\,mol^{-1}$$

Since more energy is required, why is the formation of the O^{2-} ion preferred over the formation of O^-?

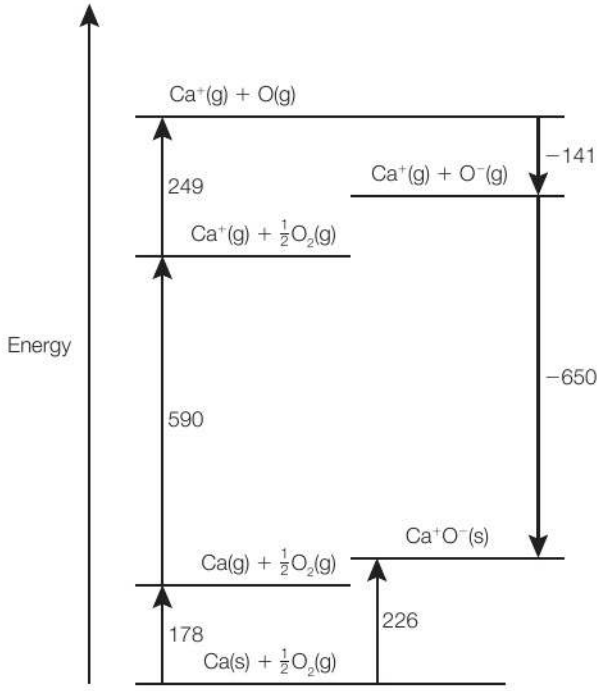
A similar pattern is observed with the formation of $Ca^+(g)$ and $Ca^{2+}(g)$ from Ca(g):

$$Ca(g) \rightarrow Ca^+(g) + e^- \qquad 1^{st}\,IE = +590\,kJ\,mol^{-1}$$

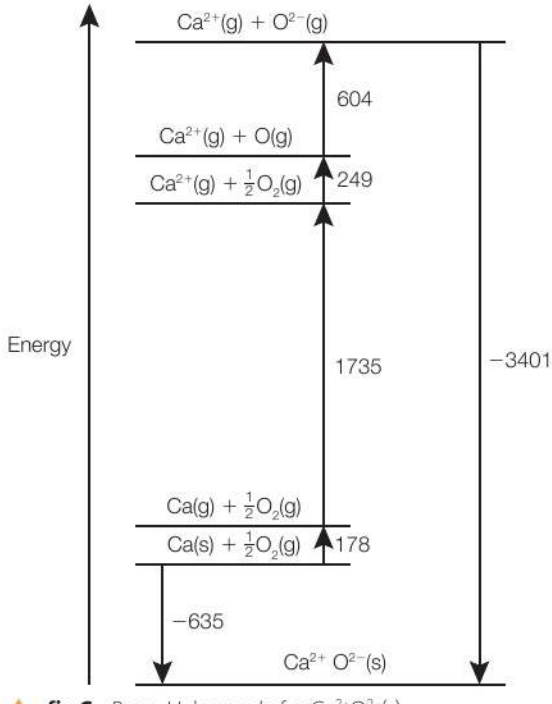
$$Ca(g) \rightarrow Ca^{2+}(g) + 2e^- \qquad 1^{st}\,IE + 2^{nd}\,IE = +1735\,kJ\,mol^{-1}$$

Similarly, since more energy is required, why is the formation of the Ca^{2+} ion preferred over the formation of the Ca^+ ion?

To answer this question it is necessary to consider *all* of the energy changes involved in the formation of an ionic compound from its elements – not just those involved in the formation of the gaseous ions from the gaseous atoms. In other words, we need to look at the information supplied by the Born–Haber cycle for each compound (see **fig B** and **fig C**).



▲ **fig B** Born–Haber cycle for $Ca^+O^-(s)$.



▲ **fig C** Born–Haber cycle for $Ca^{2+}O^{2-}(s)$.

Using the equation:

$$\Delta_f H[Ca^+O^-(s)] = \Delta_{at}H[Ca(s)] + 1^{st}\,IE[Ca(g)] + \Delta_{at}H[\tfrac{1}{2}O_2(g)] + 1^{st}\,EA[O(g)] + \Delta_{LE}H[Ca^+O^-(s)]$$

we can calculate a value for $\Delta_f H[Ca^+O^-(s)]$ using a theoretical value for $\Delta_{LE}H[Ca^+O^-(s)]$ of $-650\,kJ\,mol^{-1}$.

In this way, the value calculated for $\Delta_f H[Ca^+O^-(s)]$ is:

$$178 + 590 + 249 + (-141) + (-650) = +226\,kJ\,mol^{-1}.$$

Similarly, the value calculated for $\Delta_f H[Ca^{2+}O^{2-}(s)]$ is $-635\ kJ\ mol^{-1}$.

From these values, it is clear that the formation of $Ca^{2+}O^{2-}$ is energetically more favourable than the formation of $Ca^{+}O^{-}$.

The extra energy required to form the 2+ and 2− ions is more than compensated for by the much larger (i.e. more negative) lattice energy of $Ca^{2+}O^{2-}$ ($-3401\ kJ\ mol^{-1}$) compared to that of $Ca^{+}O^{-}$ ($-650\ kJ\ mol^{-1}$).

It is the *overall* energy change involved, and not the incorrect notion of desirability to obtain a noble gas electronic configuration, that determines how atoms and molecules will interact with one another to form a compound.

However, it is important to recognise that the conditions under which a reaction takes place will influence the overall energy change involved, and hence the nature of the compound formed.

For example, if iron is heated in chlorine gas, iron(III) chloride is the product. However, if hydrogen chloride gas is used in place of chlorine, iron(II) chloride is formed.

Therefore, changing the conditions of temperature and/or pressure can also have an effect on the exact compound formed.

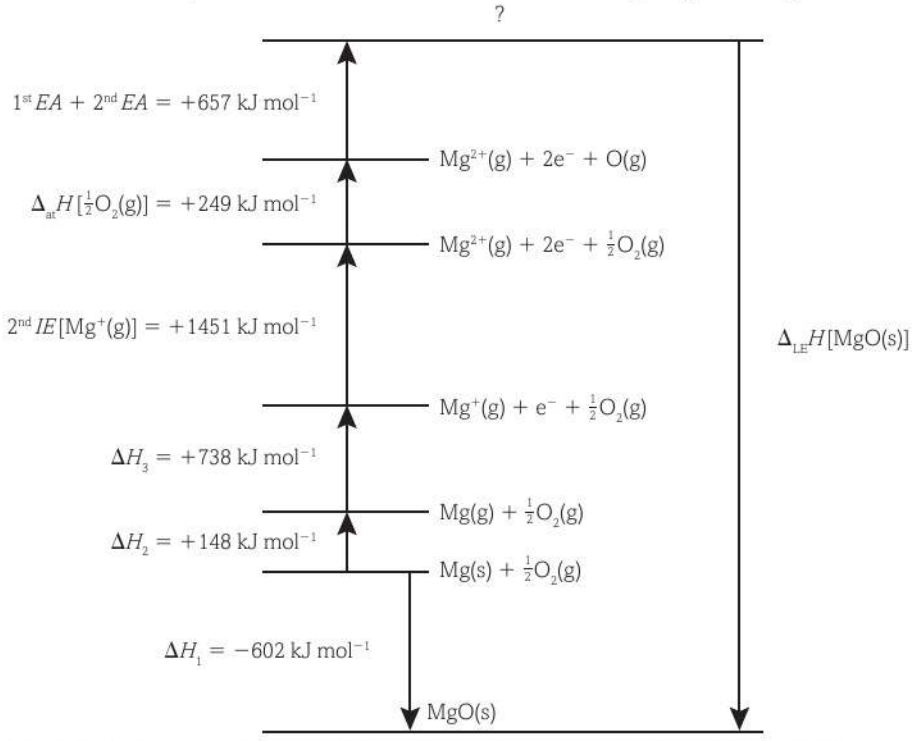
CHECKPOINT

1. (a) Explain why the lattice energy of sodium fluoride, $Na^{+}F^{-}$, is larger (i.e. more negative) than the lattice energy of potassium chloride, $K^{+}Cl^{-}$.

(b) Explain why the lattice energy of calcium oxide, $Ca^{2+}O^{2-}$, is approximately four times larger (i.e. more negative) than the lattice energy of potassium fluoride, $K^{+}F^{-}$.

[Ionic radii: Ca^{2+} 0.100 nm; K^{+} 0.138 nm; O^{2-} 0.140 nm; F^{-} 0.133 nm.]

2. The Born–Haber cycle can be used to calculate the lattice energy, $\Delta_{LE}H$, for magnesium oxide.



(a) (i) State the names for each of the enthalpy changes labelled ΔH_1, ΔH_2 and ΔH_3.

(ii) Give the formula missing at the top of the cycle, indicated by a question mark (?). Give the state symbols.

(b) The equations representing the first and second electron affinities of oxygen are:

$O(g) + e^{-} \rightarrow O^{-}(g)$ $\quad 1^{st}\ EA = -141\ kJ\ mol^{-1}$

$O^{-}(g) + e^{-} \rightarrow O^{2-}(g)$ $\quad 2^{nd}\ EA = +798\ kJ\ mol^{-1}$

Suggest why the second of these processes is endothermic.

(c) Use the information in the Born–Haber cycle shown above to calculate the lattice energy of magnesium oxide, $\Delta_{LE}H[MgO(s)]$.

(d) Explain how the lattice energy of barium oxide differs from that of magnesium oxide.

SUBJECT VOCABULARY

standard lattice energy (of an ionic compound) the energy change measured at a stated temperature, usually 298 K, and 100 kPa when one mole of an ionic compound is formed from its ions in the gaseous state

standard enthalpy change of atomisation (of an element) the enthalpy change measured at a stated temperature, usually 298 K, and 100 kPa when one mole of gaseous atoms is formed from an element in its standard state

(first) electron affinity (of an element) the energy change when each atom in a mole of atoms in the gaseous state gains an electron to form a −1 ion

SKILLS INTERPRETATION

12B 2 EXPERIMENTAL AND THEORETICAL LATTICE ENERGIES

SPECIFICATION REFERENCE
12.14 12.15

LEARNING OBJECTIVES

- Understand that a comparison of the experimental lattice energy value (obtained from a Born–Haber cycle) with the theoretical value (obtained from electrostatic theory) in a particular compound indicates the degree of covalent bonding.
- Understand that polarisation of anions by cations leads to some covalency in an ionic bond, based on evidence from the Born–Haber cycle.

EXPERIMENTAL LATTICE ENERGY

The Born–Haber cycle allows us to calculate a value for the lattice energy of an ionic compound from knowledge of the other energy changes, all of which can be determined experimentally. The value of the lattice energy calculated this way is called the **experimental lattice energy**.

LEARNING TIP

Be very careful not to confuse experimental and theoretical lattice energies.

The experimental lattice energy is calculated from a Born–Haber cycle.

The theoretical lattice energy is calculated using the principles of electrostatics.

THEORETICAL LATTICE ENERGY

The type of lattice structure and the inter-ionic distance can be found by X-ray crystallography. Using this information, it is possible for us to calculate a value for the lattice energy of an ionic compound. However, we first need to make the following assumptions.

- The ions are in contact with one another.
- The ions are perfectly spherical.
- The charge on each ion is evenly distributed around the centre so that each ion can be considered as point charges.

A value for the **theoretical lattice energy** can be calculated using the principles of electrostatics. There are three main methods for performing such calculations. If you are interested, you can research them under the headings of the 'Born–Landé equation', the 'Born–Mayer equation' and the 'Kapustinskii equation'.

Table A shows a comparison of the experimental lattice energies (obtained by using a Born–Haber cycle) with the theoretical lattice energies (calculated using the principles of electrostatics) for various compounds.

EXAM HINT

The large differences between the theoretical and experimental lattice energies for AgCl and AgBr help to explain why these compounds are insoluble in water. The difference is smaller for AgF, and AgF is soluble in water.

COMPOUND	EXPERIMENTAL LATTICE ENERGY/kJ mol^{-1}	THEORETICAL LATTICE ENERGY/kJ mol^{-1}
NaF	−918	−912
NaCl	−780	−770
NaBr	−742	−735
AgF	−958	−920
AgCl	−905	−833
AgBr	−891	−816

table A

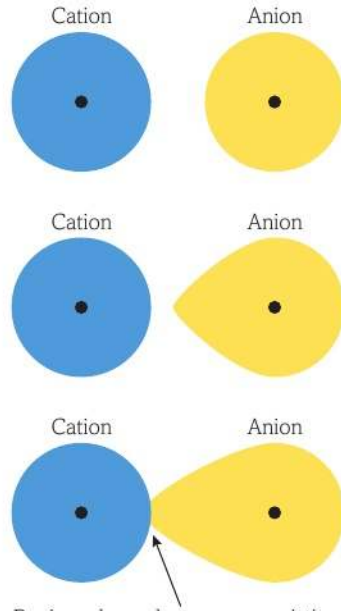
You will notice that there is good agreement between the experimental values and the theoretical values for the halides of sodium. However, the agreement is not so good for the halides of silver.

Agreement between the experimental and theoretical values of the lattice energy of a compound indicates that the ionic model is a good one for that compound. A significant difference suggests that the ionic model needs to be modified. In such compounds, the bonding in the lattice has a considerable covalent character, which makes the experimental value for the lattice energy more negative than the theoretical value.

The covalency in bonding is caused by **polarisation** of the anion by the cation. Polarisation results in distortion of the electron density within the anion, resulting in a higher electron density near the cation. This means that there is some electron density existing between the two ions. In other words, there will be a degree of covalent bonding in the compound.

EXTENT OF COVALENT CHARACTER: POLARISATION OF THE ANION

In an ionic lattice, the positive ion (cation) will attract the electrons of the anion. If the electrons are pulled towards the cation, the anion is said to be *polarised* because the even distribution of its electron density has been distorted (**fig A**).



▲ **fig A** A representation of a cation attracting the electrons of an anion in an ionic lattice.

The extent to which an anion is polarised by a cation depends on several factors. The two main factors are summarised below. These are known as Fajan's Rules.

Polarisation will be increased by:

- a high charge and small size of the cation (i.e. a high charge density of the cation)
- a high charge and large size of the anion.

HIGH CHARGE AND SMALL SIZE OF CATION

The ability of a cation to attract electrons from the anion towards itself is called its 'polarising power'. A cation with a high charge and a small radius has a large polarising power. An approximate value for the polarising power of a cation can be obtained by calculating its charge density (sometimes called surface charge density). The charge density of a cation is the charge divided by the surface area of the ion. If the ion is assumed to be a sphere, its surface area is equal to $4\pi r^2$, where r is the ionic radius.

A rough approximation of the charge density can be determined by dividing the charge by the square of its ionic radius. This calculation is beyond the scope of your International A Level course.

$$\text{charge density} \sim \frac{\text{charge}}{r^2}$$

Table B compares the extent of covalent bonding in sodium chloride and magnesium chloride.

COMPOUND	CHARGE DENSITY OF CATION	EXPERIMENTAL LATTICE ENERGY/kJ mol^{-1}	THEORETICAL LATTICE ENERGY/kJ mol^{-1}	PERCENTAGE DIFFERENCE	EXTENT OF COVALENT BONDING
NaCl	$\frac{1}{(0.095)^2} = 111$	−780	−770	1.28	very little
$MgCl_2$	$\frac{2}{(0.060)^2} = 556$	−2526	−2326	7.92	more than in NaCl

table B

The charge density of the magnesium ion is larger than that of the sodium ion, resulting in greater polarisation of the chloride ion and greater covalent bonding in $MgCl_2$ than in NaCl.

HIGH CHARGE AND LARGE SIZE OF THE ANION

The ease with which an anion is polarised depends on its charge and its size. Anions with a large charge and a large size are polarised most easily.

Table C compares the extent of covalent bonding in silver fluoride and silver iodide.

COMPOUND	CHARGE OF ANION	RADIUS OF ANION/nm	EXPERIMENTAL LATTICE ENERGY/kJ mol^{-1}	THEORETICAL LATTICE ENERGY/kJ mol^{-1}	PERCENTAGE DIFFERENCE	EXTENT OF COVALENT BONDING
AgF	−1	0.133	−958	−920	3.97	fairly large
AgI	−1	0.215	−889	−778	12.49	greater than in AgF

table C

The larger iodide ion is more easily polarised, which leads to a greater degree of covalent bonding in silver iodide.



▲ **fig B** Sodium chloride.



▲ **fig C** Silver chloride.

DID YOU KNOW?

The interesting case of silver compounds

Table D compares the extent of covalent bonding in sodium chloride and silver chloride.

COMPOUND	CHARGE DENSITY OF CATION	EXPERIMENTAL LATTICE ENERGY/ kJ mol^{-1}	THEORETICAL LATTICE ENERGY/ kJ mol^{-1}	PERCENTAGE DIFFERENCE	EXTENT OF COVALENT BONDING
NaCl	$\frac{1}{(0.095)^2} = 111$	−780	−770	0.13	very little
AgCl	$\frac{1}{(0.126)^2} = 63$	−905	−833	7.96	much greater than in NaCl

table D

Despite the polarising power of the Ag^+ ion being less than that of the Na^+ ion, there is considerably more covalent bonding in AgCl.

This is explained by considering the valence shell electronic configurations of the two ions concerned.

Na^+ $1s^2\ 2s^2\ 2p^6$

Ag^+ [Kr] $4d^{10}$

A d^{10} configuration offers less shielding than a p^6 electronic configuration, so compounds that have a d^{10} configuration show a greater tendency towards a covalent character.

For the same reason, zinc compounds also contain a significant degree of covalent bonding.

CHECKPOINT

SKILLS REASONING

1. The experimental and theoretical lattice energies, in kJ mol^{-1}, of calcium fluoride, CaF_2, and silver fluoride, AgF, are given below.

	experimental value	theoretical value
CaF_2	−2630	−2609
AgF	−958	−920

Suggest why there is good agreement between the two values for CaF_2, but there is a significant difference between the two values for AgF.

2. Anions in an ionic lattice can be polarised by the cations adjacent to them. The extent of polarisation depends on the nature of both the anion and the cation involved.

(a) Explain what is meant by the term 'polarised' in this context.

(b) State whether the oxide ion, O^{2-}, or the sulfide ion, S^{2-}, is more easily polarised.

(c) Place the following cations (with ionic radii shown in parenthesis) in order of increasing polarising power:

Mg^{2+} (0.072 nm) Al^{3+} (0.053 nm) Li^+ (0.074 nm)

Na^+ (0.102 nm) Ca^{2+} (0.100 nm) K^+ (0.138 nm)

Support your conclusion with suitable calculations.

3. Suggest why the oxide $Na^{2+}O^{2-}$ does not exist. State what further energy change, other than those quoted in the data books, you would require in order to confirm your suggestion.

SUBJECT VOCABULARY

experimental lattice energy the lattice energy calculated from a Born–Haber cycle
theoretical lattice energy the lattice energy calculated using the principles of electrostatics
polarisation the distortion of the electron density of a negative ion (anion)

12B 3 ENTHALPY CHANGES OF SOLUTION AND HYDRATION

SPECIFICATION REFERENCE
12.16 12.17 12.18

LEARNING OBJECTIVES

- Define the terms 'enthalpy change of solution, $\Delta_{sol}H$', and 'enthalpy change of hydration, $\Delta_{hyd}H$ of an ion'.
- Use energy cycles and energy level diagrams to calculate the enthalpy change of solution of an ionic compound using the enthalpy change of hydration and the lattice energy
- Understand the effect of ionic charge and ionic radius on the values of the enthalpy change of hydration and the lattice energy of an ionic compound.

ENTHALPY CHANGE OF SOLUTION, $\Delta_{sol}H$

The solubilities of ionic solids in water show a very wide variation, and there is no obvious pattern.

One of the factors that determines solubility is the value of the **enthalpy change of solution, $\Delta_{sol}H$**, the enthalpy change when one mole of an ionic solid dissolves in water to form an infinitely dilute solution.

The enthalpy change of solution for sodium chloride is the energy change associated with the following process:

$$NaCl(s) \xrightarrow{aq} Na^+(aq) + Cl^-(aq)$$

It is important to specify the extent of dilution of the final solution when quoting a value for the enthalpy change. Upon dilution, the ions in the solution move further apart (an endothermic process) and also become more hydrated (an exothermic process).

The relative importance of these two processes (endothermic and exothermic) changes with dilution, and they affect the value of $\Delta_{sol}H$ in a complicated way. For this reason, the quoted values for $\Delta_{sol}H$ refer to an **infinitely dilute solution**. This value cannot be determined experimentally and is found by a process of extrapolation.

In practice, there comes a point when further dilution has no measurable effect on $\Delta_{sol}H$. This is known as the point of infinite dilution.

Enthalpy changes of solution can be either negative or positive, as is shown in **table A**.

IONIC SOLID	EQUATION	$\Delta_{sol}H$/kJ mol^{-1}
NaCl	$NaCl(s) \xrightarrow{aq} Na^+(aq) + Cl^-(aq)$	+11.0
NaOH	$NaOH(s) \xrightarrow{aq} Na^+(aq) + OH^-(aq)$	−44.5
NH_4NO_3	$NH_4NO_3(s) \xrightarrow{aq} NH_4^+(aq) + NO_3^-(aq)$	+25.7
$MgSO_4$	$MgSO_4(s) \xrightarrow{aq} Mg^{2+}(aq) + SO_4^{2-}(aq)$	−91.3

table A

ENTHALPY CHANGE OF HYDRATION, $\Delta_{hyd}H$

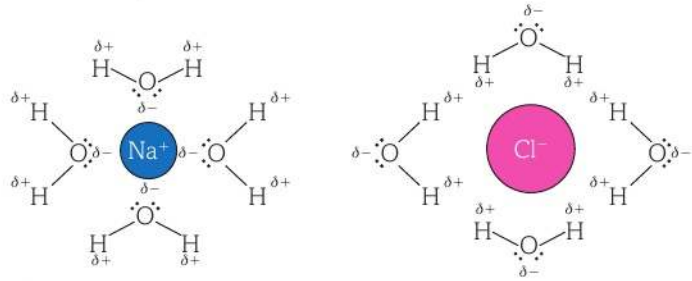
The **enthalpy change of hydration, $\Delta_{hyd}H$**, is the enthalpy change when one mole of an ion in its gaseous state is completely hydrated by water. In practice, complete hydration is said to have occurred when the solution formed is at infinite dilution.

For the sodium and chloride ions, the enthalpy change of hydration is the enthalpy change for the following processes:

$$Na^+(g) \xrightarrow{aq} Na^+(aq)$$

and $$Cl^-(g) \xrightarrow{aq} Cl^-(aq)$$

When an ion is placed in water it immediately interacts with the water molecules. Water molecules are polar (see **Topic 3 (Book 1: IAS)**) and are attracted to both positive and negative ions. **Fig A** shows the hydration of sodium and chloride ions.



▲ **fig A** Hydration of sodium and chloride ions.

In the case of the sodium ion, the interaction is the result of the attraction between the δ^- oxygen atom of the water molecule and the cation. Such an interaction is often referred to as an **ion–dipole interaction**. With some other positive ions, notably those of the transition metals, a dative covalent bond is formed between the water molecule and the cation using one of the lone pairs of electrons on the oxygen atom.

Ion–dipole interactions exist in the hydrated chloride ion, but some hydrogen bonds are also formed between the $\delta+$ hydrogen atoms of the water molecules and the chloride ion, making use of the lone pairs of electrons on the chloride ion (see **Topic 7 (Book 1: IAS)**).

Enthalpy changes of hydration are always negative. Some examples are given in **table B**.

ION	IONIC RADIUS/nm	EQUATION	$\Delta_{hyd}H$/kJ mol^{-1}
Na^+	0.102	$Na^+(g) \xrightarrow{aq} Na^+(aq)$	−406
K^+	0.138	$K^+(g) \xrightarrow{aq} K^+(aq)$	−322
Rb^+	0.149	$Rb^+(g) \xrightarrow{aq} Rb^+(aq)$	−301
Mg^{2+}	0.072	$Mg^{2+}(g) \xrightarrow{aq} Mg^{2+}(aq)$	−1920
Ca^{2+}	0.100	$Ca^{2+}(g) \xrightarrow{aq} Ca^{2+}(aq)$	−1650
Sr^{2+}	0.113	$Sr^{2+}(g) \xrightarrow{aq} Sr^{2+}(aq)$	−1480
Cl^-	0.180	$Cl^-(g) \xrightarrow{aq} Cl^-(aq)$	−363
Br^-	0.195	$Br^-(g) \xrightarrow{aq} Br^-(aq)$	−335
I^-	0.215	$I^-(g) \xrightarrow{aq} I^-(aq)$	−293

table B

EXAM HINT

When you **write** an equation for the enthalpy change of hydration of an ion, make sure that you put the gaseous ion on the left-hand side.

LEARNING TIP

It is important to understand what is meant by 'magnitude' here. A value for a hydration enthalpy of −322 is smaller than a value of −406. The negative sign refers to the type of enthalpy change; in this case, it is an exothermic change. Less energy is given out if the change is −322 than if the change is −406. So −322 has a smaller magnitude than −406.

In mathematics, the negative sign could indicate a negative number, in which case −322 would be greater than −406.

We can also describe the trend in hydration enthalpies by saying that the values become less negative as we go down a group.

FACTORS AFFECTING THE MAGNITUDE OF THE HYDRATION ENTHALPY

The first thing you will notice from the values in **table B** is that $\Delta_{hyd}H$ is much more negative for 2+ ions than for 1+ ions. This makes sense because we would expect a doubly charged ion to have a stronger interaction with the water molecules compared to a singly charged ion. That is to say, the electrostatic force of attraction between a doubly charged ion and water molecules will be greater than that between a singly charged ion and water molecules.

As we go down a group (e.g. Na^+ to Rb^+ or Cl^- to I^-) the magnitude of $\Delta_{hyd}H$ decreases. This seems to correlate with an increase in ionic radius. This trend can be explained using a simple electrostatic model for hydration, similar to that used to explain the variation in lattice energies for ionic solids (see **Section 12B.1**). As the ions become larger, the electrostatic force of attraction between them and the water molecules decreases, and hence the energy released upon hydration decreases.

As with lattice energy, there is a strong correlation between $\Delta_{hyd}H$ and the charge densities of the ions. **Table C** shows this for four cations.

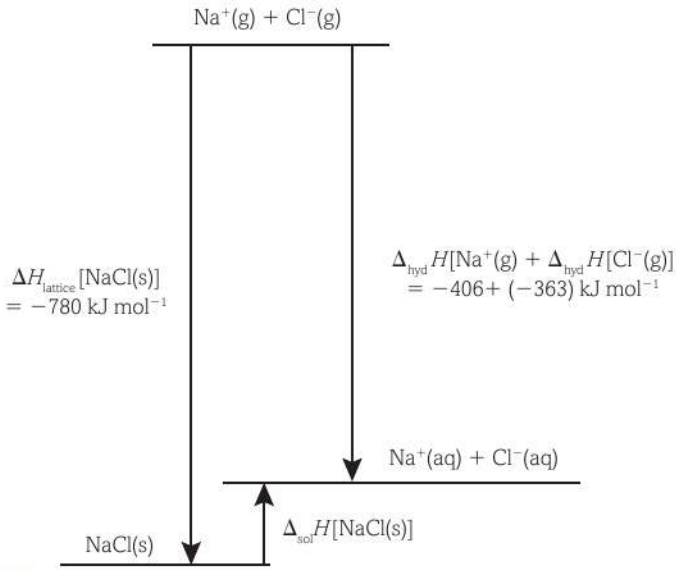
CATION	CHARGE DENSITY OF CATION	$\Delta_{hyd}H$/kJ mol^{-1}
Na^+	$\frac{1}{(0.095)^2} = 111$	−406
K^+	$\frac{1}{(0.133)^2} = 57$	−322
Mg^{2+}	$\frac{2}{(0.060)^2} = 556$	−1920
Ca^{2+}	$\frac{2}{(0.099)^2} = 204$	−1650

table C

The greater the charge density of the cation, the more negative the value of $\Delta_{hyd}H$.

RELATIONSHIP BETWEEN $\Delta_{sol}H$, $\Delta_{hyd}H$ AND $\Delta_{LE}H$

The relationship between $\Delta_{sol}H$, $\Delta_{hyd}H$ and $\Delta_{LE}H$ is best shown by an energy level diagram similar to that of a Born–Haber cycle as shown in **fig B**.



▲ **fig B** Energy level diagram for the dissolution of sodium chloride.

Applying Hess's Law:

$$\Delta_{LE}H[NaCl(s)] + \Delta_{sol}H[NaCl(s)] = \Delta_{hyd}H[Na^+(g)] + \Delta_{hyd}H[Cl^-(g)]$$

$$\Delta_{sol}H[NaCl(s)] = -406 + (-363)\,kJ\,mol^{-1} - (-780)\,kJ\,mol^{-1}$$

$$= +11\,kJ\,mol^{-1}$$

The relationship can also be shown in the form of a Hess cycle (shown in question 3 below).

CHECKPOINT

SKILLS INTERPRETATION

1. When potassium fluoride dissolves in water, the lattice breaks up and the potassium and fluoride ions become hydrated.
(a) Draw diagrams to represent (i) a hydrated potassium ion and (ii) a hydrated fluoride ion.
(b) Name the type of interaction that occurs between the water molecules and the ion for both the hydrated potassium ion and the fluoride ion. Describe how each interaction occurs.

2. These data refer to some of the energy changes involved when magnesium chloride dissolves in water.

$$Mg^{2+}(Cl^-)_2(s) \xrightarrow{aq} Mg^{2+}(aq) + 2Cl^-(aq)$$

$$\Delta_{sol}H[Mg^{2+}(Cl^-)_2(s)] = -155\,kJ\,mol^{-1}$$

$$Mg^{2+}(g) + 2Cl^-(g) \rightarrow Mg^{2+}(Cl^-)_2(s)$$

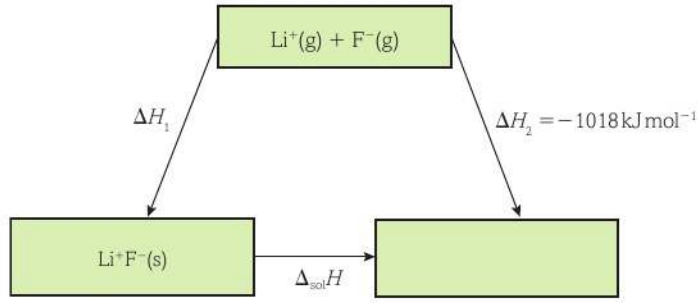
$$\Delta_{LE}H[Mg^{2+}(Cl^-)_2(s)] = -2526\,kJ\,mol^{-1}$$

$$Mg^{2+}(g) \xrightarrow{aq} Mg^{2+}(aq)$$

$$\Delta_{hyd}H[Mg^{2+}(g)] = -1920\,kJ\,mol^{-1}$$

Use this information to calculate the enthalpy change of hydration of the chloride ion, $\Delta_{hyd}H[Cl^-(g)]$.

3. The diagram below is a Hess cycle for the dissolving of lithium fluoride in water.



(a) Complete the cycle by filling in the empty box.
(b) State the name of the energy change represented by ΔH_1.
(c) Apply Hess's Law to obtain an expression for $\Delta_{sol}H$ in terms of ΔH_1 and ΔH_2.
(d) Calculate $\Delta_{sol}H[Li^+F^-(s)]$, given that $\Delta H_1 = -1031$ kJ mol^{-1}.

4. The standard enthalpy change of solution of sodium fluoride is +0.3 kJ mol^{-1}.

A sample of sodium fluoride of mass 1 g is added to 250 cm^3 of water in a beaker and stirred with a thermometer graduated in intervals of 1 °C.

Explain what is likely to happen to the reading on the thermometer as the sodium fluoride dissolves. No calculation is necessary.

SUBJECT VOCABULARY

enthalpy change of solution, $\Delta_{sol}H$ the enthalpy change when one mole of an ionic solid dissolves in water to form an infinitely dilute solution
infinitely dilute solution a solution in which there is so much water that adding any more does not cause a further enthalpy change
enthalpy change of hydration, $\Delta_{hyd}H$ is the enthalpy change when one mole of an ion in its gaseous state is completely hydrated by water
ion–dipole interaction the attraction in aqueous solution between ions and polar water molecules